HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics 2

2021/22 Semester 2

SOLUTIONS

Question 1

(a) $\mu_{SS:SO} = \mu_{SS} + \mu_{SO} = 0.000976 + 0.000507$

$= 0.001483$ [1]

(b) $\left(\frac{L^{m+t}_{60}}{L^{t}_{50}}\right)\left(1 - \frac{L^{m}_{70}}{L^{m}_{60}}\right) + \left(1 - \frac{L^{t}_{60}}{L^{t}_{50}}\right)\left(\frac{L^{m}_{70}}{L^{m}_{60}}\right)$ [1]

$= \left(\frac{9848.4}{9952.7}\right)\left(1 - \frac{9258.1}{9826.1}\right)$

$+ \left(1 - \frac{9848.4}{9952.7}\right)\left(\frac{9258.1}{9826.1}\right)$

$= 0.0691$ [1]

(c) $a^{(4)}_{SS|SO} \approx \ddot{a}_{SS|SO}$ [1]

$= \ddot{a}^{m}_{SO} - \ddot{a}^{t+m}_{SS:SO}$

$= 18.843 - 16.909$

$= 1.934$ [1]

Question 2

(a) The expected present value of £1 paid immediately on death of a life (x) if it dies before the life (y) and within 10 years. [1]

Notation: $\bar{A}^{1}_{xy:\overline{10}|}$ [1]

(b) The probability that (x) dies second and before time 20. [1]

Notation: $\mathcal{L}^{2}_{xy:\overline{20}|}$ [1]

or $_{20}q^{2}_{xy}$

Question 3

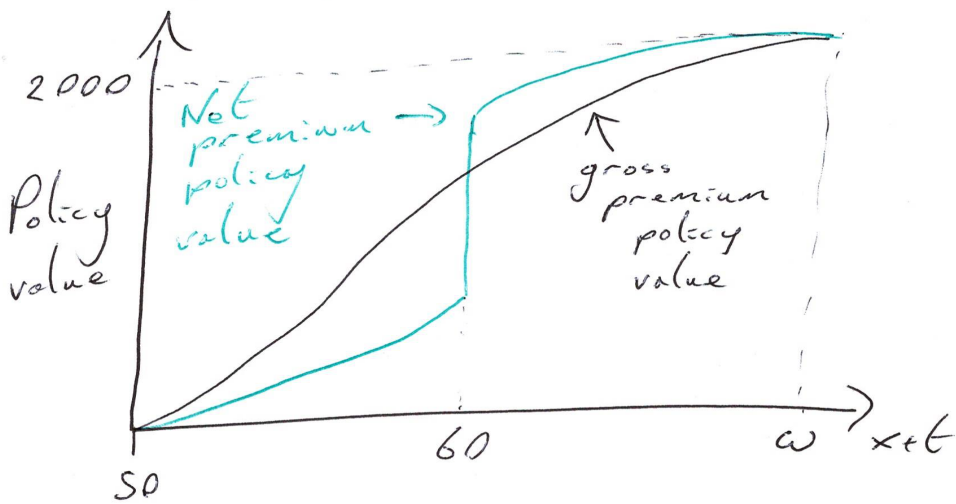
(a) $\frac{d}{dt}V(t) = V(t)\delta + \bar{P} - \mu_{x+t}(1000 - V(t))$ [1]

(b) [1] mark for GPPV and NPPV both equal to zero at time zero

[1] mark for NPPV jumps up at time 10 (when bonus declared and GPPV smooth.

[1] mark for NPPV > GPPV after time 10 (since EPV(outgo) is the same for both after time 10 but net premiums are lower than gross in this case).

Example of an acceptable answer:



(c) V(t) is not smooth between integer ages/times. It jumps up each time a premium is paid. We need to capture these jumps, which linear interpolation does not but recursive relationships do. [1]

(d) They can offer a relatively low initial guaranteed sum assured which they are confident of being able to pay even if mortality is high and investment returns very low. Then increase the SA as term to expiry reduces + risk reduces. [1]

Question 4

(a) Calculate the standardised mortality ratio (SMR) for Policy 1 and Policy 2 compared to the standard population.

*SMR = (actual deaths) / (sum of the standard death rate * Policy population in each age band)* [1 mark]

SMR policy 1 = 93%
SMR policy 2 = 113% [1 mark]

(b) Explain why the crude death rate (CDR) is the same for both policies even though death rates have been higher every year under Policy 2.

There are lots of lapses from policy 2, so the average age of the population is lower. Hence more of the exposure is in the age bands where mortality is not so high.
[1 mark for lower average age; 1 mark for mentioning lapses or surrenders]

(c) List, with brief reasons, 3 types of selection which appear to be present in the table above and explain what might have caused each of them.

No marks for answers without reasons. 2 marks for each type with a reason/cause which is fully convincing. 1 mark if the reason is not so convincing. Any type is acceptable if there is a good argument. In particular:

- *There are a lot of lapses in policy 2. This is only possible because it is a level premium, not single premium (like policy 1). The higher mortality in policy 2 vs policy 1 might be caused by selective decrements because those that lapse might be heathier than those that don't.*

- *There seems to be temporary initial selection in policy 1 with very low mortality in the first year. This could be caused by the rigorous underwriting in policy one.*

- *The higher mortality in policy 2 could be caused by class selection. Eg, the lower sum assured might mean it is bought by a higher-risk group.*

[up to 6 marks]

[Total 10 marks]

Question 5

(a) Calculate the profit vector and profit signature for this policy.

1 mark for each of the highlighted cells or columns

Unit fund

Year (t)	Premium	Allocation Factor	B\O spread	Premium Allocated	Policy Fee	Fund return	Fund before invt return	Invt return	Fund before FMC	FMC	Fund after FMC	Mortality charge	Year-end Fund
1	£1,000.00	0.95	2%	931.00	0.00	3%	931.00	27.93	958.9	24.8	934.17	0.00	934
2	£1,000.00	1	2%	980.00	0.00	5%	1914.17	95.71	2009.9	59.3	1950.53	0.00	1951
3	£1,000.00	1.01	2%	989.80	0.00	10%	2940.33	294.03	3234.4	123.5	3110.87	0.00	3111

Non-unit (sterling) fund

Year (t)	Unallocated Premium	Expense	Policy Fee	Interest	FMC	Fund management Expenses	MC	q (ELT15 females)	Expected Death Cost	SFC_t (=profit vector)	$_{t-1}p_x$	Profit signature
1	69.00	100.00	0.00	-0.62	24.76	14.38	0.00	0.00219	2.19	-23.43	1.000000	-23.43
2	20.00	10.00	0.00	0.20	59.34	30.15	0.00	0.00241	2.41	36.98	0.997810	36.90
3	10.20	10.00	0.00	0.00	123.49	48.52	0.00	0.00266	2.66	72.52	0.995405	72.19

Common errors on this question included:

(1) Not projecting the unit fund and non-unit fund separately and failing to understand that the non-unit fund is the insurer's cashflows, hence profit/loss

(2) Getting mixed up between charges/fees (amounts like the FMC paid by the member to the insurer, ie cashflows from the unit fund to the non-unit fund) and costs/expenses (actual amounts paid by the insurer to cover its costs, which we make an assumption for but may be unpredictable in practice)

(3) There is no mortality charge. The expected death cost is just $1000q_{x+(t-1)}$. Note that the death benefit is £1000 plus the bid value of the units, but the bid value of the units is just paid from the unit fund, so the insurer only has to finance £1000 on death.

(4) The FMC is calculated in year 1 as 27.93*20% + 958.9*2% = 24.8 (and similarly in years 2 and 3).
(5) Applying interest to the fund management expenses or FMC in the non-unit fund. This is incorrect because these cashflows occur at year end.
(6) Sign errors. Eg, the FMC is a negative cashflow in the unit fund and positive in the non-unit fund.

[8 marks]

(b) Suggest, with reasons, which feature of this policy is most likely to result in profit being higher or lower than expected.

No marks for answers without reasons. 2 marks for a suggestion with a reason which is fully convincing. 1 mark if the reason is not so convincing. Expected answer:

"Since premiums are fixed (hence we know what we're getting from unallocated premiums), the main uncertainty is how much income we will receive from the FMC vs fund management expenses. The FMC is structured to reward successful active management and can be very high (profitable) if there are good investment returns or low if there are not."

[2 marks]

(c) Calculate the profit signature allowing for surrenders.

1 mark for each of the highlighted cells or columns

Year	**SFC_t (=profit vector)**	**sigma_t**	**surrenders (as % policies in force at year start)**	**surrender penalty (of bid value)**	**surrender penalty (£)**	**surrender penalty expected value (per policy in force at t-1)**	**survival allowing for surrender**	**sigma_t allowing for surrenders**
1	-23.43	-23.43	20%	10%	93.42	18.68	1.00000	-4.75
2	36.98	36.90	50%	5%	97.53	48.76	0.79825	68.45
3	72.52	72.19	0%	0%	0.00	0.00	0.39816	28.88

The final column can be calculated as, eg, 0.79825*(48.76 + 36.98) = 68.45

[4 marks]

[Total: 14 marks]

Question 6

(a) Differences in risk between DC and DB for a member:

* In a DB scheme the benefits are determined by a formula. The employer takes the risk of needing to pay additional contributions if benefits cost more than estimated.
* In a DC scheme, the benefits are whatever the accumulated contributions can buy. So the member carries the investment risk and (at up to the point of buying an annuity, if they choose to) mortality risk. [1 mark for saying in DC unlike DB most risk falls on the member; 1 mark for a reasonable explanation of why.

(b)

Age pension liability

$$= \frac{850}{1000}\,\frac{10}{60}\left(1.03^{(69-55)}\,60000\right)1.04^{-(60-55)}\,\ddot{a}^{(12)}_{60}$$

$$= 127{,}334.6$$

using $\ddot{a}^{(12)}_{60} = 16.652 - \frac{11}{24}$

Withdrawal liability

$$= \frac{50}{1000}\,\frac{10}{60}\left(60000\,\frac{(1.03^2 + 1.03^3)}{2}\,1.02^{11}\right)1.04^{-5}\,\ddot{a}^{(12)}_{60}$$

$$+ \frac{100}{1000}\,\frac{10}{60}\left(60000\,\frac{1.03^3 + 1.03^4}{2}\,1.02^{9}\right)1.04^{-5}\,\ddot{a}^{(12)}_{60}$$

$$= 22{,}071.9$$

Total PSL $= 127334.6 + 22071.9$

$= 149406$

In part (b), one mark for each of the following:

* Use n=10 and divide by 60 throughout
* Discount by 5 years at 4% throughout
* Use the correct annuity at age 60
* Add age and withdrawal calculations
* Age: Use the correct 850/1000 decrement
* Age: Use the correct 1.03^4 salary increase and £60k
* Withdrawal: Use the correct 50/1000 and 100/1000 decrements
* Withdrawal: Use the correct 0.5*(1.03^3 + 1.03^4) and 0.5*(1.03^2 + 1.03^3) salary increases with £60k
* Withdrawal: Use the correct revaluation in deferment

Total marks for question 6: 11 marks.

Question 7

(a) Calculate the profit signature and expected present value of profit with allowance for reserves.

[1 mark for each of the highlighted cells or columns]

Time (t)	${}_tp_x$	Cashflows at time t before allowing for reserves	Profit signature with no allowance for reserves	Reserve at time t-1	Interest on reserve	Increase in reserve	PRO(t)	Profit signature Sigma(t)
1	0.99	-100	-100	0	0.00	0	-100	-100
2	0.99	50	50	0	0.00	0	50	50
3	0.95	100	98	0	0.00	475	-375	-368
4	0.95	500	466	500	15.00	-25	540	503
5	0.95	-500	-442	500	15.00	-500	15	13

EPV profit = $\sum_t \text{sigma(t)}1.12^t = £15.63$ [1 mark]

(b) Without doing any further calculations, say what the expected present value of profit would be if the interest on reserves is 12% instead of 3%. Give reasons for your answer.

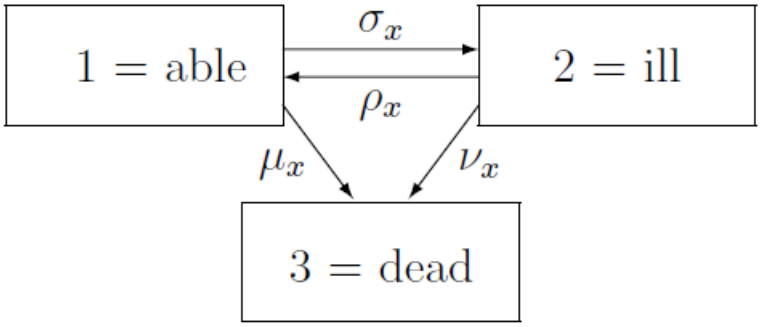
No marks for answers without reasons. 2 marks for the correct answer with a reason which is fully convincing. 1 mark if the reason is not so convincing. Expected answer:

- It is £64.84. The answer before allowing for reserves is unchanged because it isn't affected by the new assumption for interest on reserves. Now that we know reserves earn the same rate of interest as we require in the RDR, there is no cost of capital from holding reserves, so holding reserves gives the same answer, £64.84, as not holding them. (We have shown in the course reading that holding reserves doesn't affect EPV of profit if RDR=interest on reserves.)

[Total: 6 marks]

Question 8

An insurer offers 5-year Income Protection Insurance (IPI) policies. The insurer uses the following simple Markov model to price its policies.



Describe in words what each of the following integrals represents.

(a) $50{,}000 \int_1^{\infty} {}_tp_x^{\overline{11}} \mu_{x+t} e^{-\delta t} dt$

One half mark per point:

- *It is the expected present value of 50,000 paid immediately if the life dies*
- *From the able state (ie, not the ill-state)*
- *At any time after the end of a one-year waiting period*
- *If the life has never been in the ill state*

(b) $10{,}000 \int_0^{65-x} {}_tp_x^{\overline{11}} \sigma_{x+t} e^{-\delta t} \int_0^{65-(x+t)} {}_sp_{x+t}^{\overline{22}} e^{-\delta s} ds\, dt$

One half mark per point:

- *It is the expected present value of 10,000 paid continuously*
- *While in the ill state*
- *If the life has never previously been in the ill state*
- *With the policy ending at age 65*

[Total: 4 marks]